Matheism: All Is Math

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Abstract

Reality is complete mathematical structure, because everything that exists is ruled or random, and both are mathematical. A proposed nonmathematical ingredient would need a nature and definite relations to what it affects, and stating those adds more structure; the same argument covers consciousness, and since a structure is one wherever it occurs, the bare subject of experience is one self in every mind. The claim is necessary, as 2 + 2 = 4 is, and it still has teeth: it rules out every rival that needs something with no structure, and one coherent feature that is neither ruled nor random would refute it. The measure follows from the same dichotomy, and it is the part that meets data. Each bit of a shortest description is an unruled 50/50 choice, so each added bit halves a description's weight, and the choice whether to add another bit at all makes longer descriptions lose in total as well. The universe therefore starts simple, laws and starting state together, and observation agrees whatever the reason. Minds like ours are complex, so complexity has to grow from that start in causal steps, discrete on the view taken here, since a continuum has no next state. Either way, time runs forward: an endless history has no last moment to run back from, and memories are built by steps already taken. Both points are a priori, from what a step and an endless history are. Applied to quantum theory, the same measure favors evolution without collapse and one common rule for measurement settings and outcomes.

Keywords: mathematical monism, rule-random dichotomy, Kolmogorov complexity, structural realism, Simplicity Dominance Principle

Matheism holds that reality is mathematical structure, because everything that exists is ruled or random and both are mathematical. It is mathematical monism (the view that reality is one kind of thing, mathematical structure), expressed here as *Deus sive Mathematica*, God or mathematics, after Baruch Spinoza's *Deus sive Natura*, God or Nature, with mathematical structure in the role of his one necessary substance. It is not Platonism, which is dualism. Athens' early thinkers sought reason inside nature, and the Pythagoreans held that things are numbers. Plato, late in that line, set the Forms outside nature, so the Forms, the world or the link between them lies beyond reason, and reality cannot be known. Both halves turn supernatural, the Forms by standing outside nature and nature by following no logic without them. That is faith posing as reason, a late Athenian anti-philosophy, and it fails a priori: outside logic is illogical, as impossible as a square circle, and the view cannot even be stated without the logic it denies. Section 1 states the identity argument this paper uses.

Three arguments carry the account. First, a complete account of a thing's properties and relations leaves no substrate (underlying stuff that bears the properties) to add. Second, nothing further is needed for a complete structure to exist, so every consistent structure, one free of contradiction, exists. Third, swapping the supposed owners of two lives would change no experience, no memory and no relation, so a separate owner does no work. What remains in each life is the bare form of being the one who feels, and on mathematical monism a structure is one wherever it occurs, as the number two is one number however many pairs there are. That gives the one-self account, close to the open individualism of Daniel Kolak and Arnold Zuboff (Kolak, 2004; Zuboff, 2023), with distinct memories and distinct lives left in place. Each later step states its own

premises where it is used.

The measure follows from the same dichotomy. Each bit of a shortest description is an unruled 50/50 choice, so each added bit halves a description's weight, and since whether to add another bit is a second 50/50, longer descriptions lose in total as well. So the universe starts simple and has to grow complex, because minds like ours are complex. Growth takes causal steps, discrete on the view taken here, since a cause produces its effect as the next state and a continuum has none, an a priori proof in the logician's sense. Others allow continuous causes, as Russell did in holding that motion is only being at different places at different times (Russell, 1903). A continuum of positions with nothing to produce the next is the emperor's new points. The rest holds either way. Time is those steps, running forward from the simple start, since an endless chain has no last step to run back from and a reversal would have to pay for its turning point. The claim is necessary, as 2 + 2 = 4 is, and it still has teeth: it rules out every rival that needs something with no structure, one coherent feature that is neither ruled nor random would refute it, and the measure is the part that meets data.

1. The Argument

1.1 Existence requires determinacy

To exist is to be some way rather than another, and every such way is a relation that can be stated mathematically. For something to exist, it must have a nature: it must differ from nothing in some respect. A property, a possible change or an effect on something else supplies relations that can be stated mathematically. The question is what, if anything, remains once all of those features are specified, and Matheism's answer is that the complete structure is the thing itself.

Physics keeps finding relations where it looked for stuff. Each property is defined by its connections to others: mass relates to energy through E=mc^2, and position relates to velocity through derivatives, which measure rates of change. Particles became field excitations (ripples in fields that fill space), forces became geometric curvature, and the Standard Model, the established theory of elementary particles, is relational throughout. The electron is not a tiny ball with properties attached; it is a pattern of relationships, its electromagnetic coupling (how strongly it interacts with electric and magnetic fields), its responses to measurement and its exclusion-principle relations (the rule that no two electrons can occupy the same state). Remove these relations and nothing remains.

1.2 The dichotomy

Everything determinate is either ruled or random, and both are mathematical. Anything determinate can be written down as a record, such as a run of measurements, a sequence of states or the digits of a number, and the test is whether something much shorter than the record can produce all of it. If something can, the record is ruled: the laws of physics, the digits of pi and fractal curves are all ruled. If nothing much shorter can produce the record, as with a long run of fair coin flips, it is random. A law such as x = y shows both sides. It is the shorter something, three symbols that settle every case, so a record that follows it is ruled however long it grows. The law itself is a short choice: nothing shorter produces it, and no rule selects it from all the expressions that could be written, so in that sense it is random. Every consistent expression exists as a structure, and the structure is its physics. Ruled consequences unfold from unruled starting choices, and mathematics itself is built that way.

Arithmetic has its own big bang. Giuseppe Peano's axioms for the counting numbers (Peano, 1889), in their usual form, are five: 0 is a number; every number

has a next one, its successor; 0 is the successor of no number; different numbers have different successors; and whatever holds for 0, and passes from each number to the next, holds for every number. Add the laws of logic, among them that everything equals itself, that equals can stand in for each other, and that from a statement and "if it, then another," the other follows. Let 1 be the successor of 0 and 2 the successor of 1, and let adding 0 change nothing and adding a successor give the successor of the sum. Then 1 + 1 = 2, because 1 plus the successor of 0 is the successor of 1 plus 0, which is the successor of 1, which is 2. The same few lines show that 0 is not 1, since 1 is a successor and 0 is not; that the numbers do not end or circle back, since a repeat, once equal successors are canceled, would make 0 a successor; and that x = y gives y = x, since x = x holds and y can stand in for the first x. Every sum, every prime and every theorem of arithmetic is already in those lines, unfolding by rule from choices that nothing shorter decides. In symbols, writing s(n) for the successor of n, with 1 = s(0), 2 = s(1), n + 0 = n and n + s(m) = s(n + m): 1 + 1 = 1 + s(0) = s(1 + 0) = s(1) = 2.

Stated precisely, a string S, a sequence of symbols, is ruled when its Kolmogorov complexity K(S), the length of its shortest description, falls well below its own length, and random when K(S) is about the length of S. Fix the description language U and a deficiency threshold c, a set margin in bits: either K_U(s) < |s| - c or K_U(s) >= |s| - c, and there is no third option. Every string no longer than c lands in the second class, so a short law or axiom counts as random while the long record it generates counts as ruled. No formal system, a fixed set of axioms and rules for deriving theorems, can certify incompressibility beyond a fixed length (Chaitin, 1974), and the partition needs no certificate: every string falls on one side whether or not anyone can prove which. Randomness remains mathematical: an incompressible string has definite entries, and a stochastic law specifies the probabilities of its outcomes.

The dichotomy reaches everything, because being determinate means being one way rather than another, and that difference can be written down. Any proposed nonmathematical substance must have determinate features, or it is nothing. Those features are ruled or random, both are mathematical, and so the proposed substance is mathematical. The challenge to a critic is to say what a nonstructural remainder would contribute.

Chance is mathematically coherent, so the dichotomy does not require determinism. A probability law specifies outcomes and weights without fixing one outcome. Demanding sufficient conditions that determine every event is a further premise, not a result of the dichotomy, and unpredictability, algorithmic randomness and fundamental chance are distinct.

1.3 Interaction requires shared structure

Two things can act on each other only through a definite relation, so a soul with no structure could not move a body. A conversation shows why. Words arrive as sound waves and change what the listener feels, and what the listener feels moves the listener's mouth to send sound waves back: physics in, physics out. A consciousness in an ineffable dimension beyond mathematics, beyond words, would need a place where the signal leaves the mathematics and a way back in. A theory in which mind changes bodily events must specify how that relation works. The interaction argument alone does not show that everything related by one law is one substance; Section 1.4 supplies that step.

The argument applies again at every level. If a proposed extra property Q affects the physical world, its coupling has determinate features belonging to the structural account. If Q changes that account, the change must be specified; if it changes nothing, its necessity must be explained.

1.4 The identity argument

A physical system and its complete mathematical structure share every relational property, so nothing is left to tell them apart. What could distinguish them? Not their relationships, which are identical by isomorphism, a one-to-one match that preserves every relation. Not their causal powers, which supervene on (that is, are fixed by) relational structure. Not their observable properties: color depends on spectra (the mix of wavelengths in light) and perceptual organization, hardness on bonding and temperature on thermal state, and all of these reduce to relational structure.

The last candidate is a nonrelational substrate, primitive stuff bearing properties without being made of them, and it fails the same test. Any proposed substrate must itself have determinate features, or it is nothing. Those features are ruled or random, both are mathematical, and so the substrate is structure. Two systems with identical relational structure behave identically in every observable way, so the substrate does no explanatory work.

Chess shows the identity. The game can be played with carved wood, in computer memory or in pure abstraction, and no chess property differs across them: the game is the relational structure, and the wood is one way of accessing it. Physical reality and its complete mathematical structure share all constitutive properties, so they are identical. The math is the territory.

1.5 Transcendental constraint

Observation itself needs structured relations: a record must track what it records, and a learned cue must bear on the event it predicts. This does not require every observed object, or the observer's stored data, to be compressible; the constraint concerns the relations that support observation.

1.6 Mathematical structure is necessary

Mathematical structure cannot fail to exist, so the question of why there is something rather than nothing ends at necessity. Mathematical structure is uncreated: no agent makes 2+2=4, and none could make it otherwise. Causation is a relation within structure, so asking what caused structure asks for a relation outside all relations. Stephen Hawking asked what breathes fire into the equations and makes a universe for them to describe (Hawking, 1988). The math rules themselves do: applying a rule to a state is a relation within the structure, so the fire needs no source outside it.

Only consistent mathematics forms structure. Inconsistent statements can be written but not built, so consistency is the filter on what exists.

2. All Consistent Structures Exist

Every consistent structure exists as mathematics, since that takes nothing beyond consistency; whatever let one exist and not another would itself be structure, so the point is a priori. Physics is transitions in mathematics, discrete since a cause needs a next state, and consciousness is physics too. Mathematical existence already supplies the complete structure. This paper rejects a primitive actuality property that would select some existing structures and exclude others: any such property would either enter determinate relations with the structure, and so belong to it, or make no difference to anything.

Matheism reaches structural identity by a different route from Max Tegmark's Mathematical Universe Hypothesis (Tegmark, 2008). Tegmark starts from an external reality whose complete description must be free of human concepts; Matheism starts from the rule-random dichotomy and the idleness of any substrate. Every consistent law structure exists in its own right, which does not place every structure inside our particular universe. The ontological argument

concerns what existence is, and the measure concerns which structures and observations receive weight.

3. The Measure

Each bit of a shortest description is an unruled 50/50 choice, so each added bit halves a description's weight. A shortest description cannot be compressed, since its compressed version plus an instruction to unpack it would be shorter still, so its bits are random in the sense of the dichotomy. With no rule to favor 0 or 1, each value has probability 1/2, and any lean toward one value would itself be a rule written into the description. Stated precisely, fix a description language and a class of complete candidate models and assign code weights w(H) proportional to 2^(-L(H)); a real difference of Delta L bits then gives a prior ratio (the odds before any evidence) of 2^(Delta L), and ten bits give exactly 1,024. Evidence multiplies those prior odds by its likelihood ratio, how strongly it favors one account over the other.

Longer descriptions lose in total as well, because whether to add another bit is also an unruled 50/50. Each description extends by a 0 or a 1, each extension carrying half its parent's weight, so if every string counted as a complete description, every length would carry the full weight again and the total would be infinite. With a 50/50 choice to stop or go on, all descriptions one bit longer weigh half as much together, and everything longer than a given length adds up to that length's weight: the tail equals the peak. Added laws more often break the conditions observers need than create them, so at least half the weight sits at the shortest length that can produce observers. This weighting, under which the shortest descriptions that work dominate, is the Simplicity Dominance Principle.

Ray Solomonoff introduced algorithmic probability, the weighting of hypotheses by the length of the programs that produce them, for induction, reasoning from

past observations to predictions (Solomonoff, 1964), and Jürgen Schmidhuber applied it to whole universes (Schmidhuber, 2000). A physical prediction states its coding domain, its multiplicities and its conditioning rule. The simplest programs may generate no observers, so conditioning on evidence changes the comparison.

A measured parameter can be an output of a simpler rule, and then it costs nothing extra. It can also be an address within the space that rule generates. If one rule derives several constants, the rule and any remaining selector replace their separate encodings, and deriving those relations and showing the saving is a concrete route to strengthening the Simplicity Dominance Principle.

4. Core Consequences

4.1 What the simplicity prior establishes

Simplicity sets the starting odds, and evidence then decides. Under a specified code prior, shorter descriptions receive greater individual prior weight, and that weighting does not guarantee that the shortest candidate is true or best supported once the evidence is in.

4.2 The arrow of time

Time is the chain of causal steps by which a simple start grows complex minds. The measure favors a simple start, laws and starting conditions together, and observation shows the same start whatever explains it: the laws we know fit on a page, and the early universe was smooth to about one part in a hundred thousand, with no structures yet, let alone minds (Planck Collaboration, 2020). Each later step needs that fact, not its reason, so a reason can fail without taking the rest of the chain with it. Minds like ours are complex, so their complexity has

to grow from that start, as many simple rules grow complexity: the Game of Life's one rule about neighbors grows five live cells into more than a thousand generations of structure and can run any computation (Gardner, 1970). Growth takes steps, discrete on the view taken here, because a cause produces its effect as the next state and a continuum has no next state for a cause to produce; a reader who allows continuous or uncaused change still has growth from a simple start, which is all the arrow needs. A process generating each event from the previous one needs a starting state, and an endless one has no final event to work back from, so a complete backward production has nowhere to begin. Each causal step is a rule applied to rules: the rules are the only things, and a structure that only sat there would still exist, as mathematics, and know nothing, like a dead brain. Time is the applying, and space is how far it spreads, at the base layer's rate, the speed of light. Light, which crosses space while no time passes for it, is no exception: a moving body's transitions run slower, and light spends every step moving, so nothing within it could move relative to the rest.

Reversing a finite history avoids the missing endpoint, and pays in weight for whatever extra it needs. The first extra is the turning point: the endless forward history does not have to say where it stops, while a reversal has to say where it starts back from, and each bit of that address halves its weight again. Specifying the turning step takes about as many bits as writing its number in binary: about 20 bits for a typical step near one million, which leaves about a millionth of the weight, so later turning points cost more. A code can charge a finished string and its reversal the same, but only by running the forward history and reversing the output, so the reversed history still depends on the forward production.

Minds run forward too. A thought merges many possible states into one, so only the forward step is produced by the mind's own present state; run backward, its next state would be set by heat converging from its surroundings. Memories are

built by steps already taken, which is why observers remember the past and not the future.

4.3 What observer selection explains

We see constants that allow observers because only such constants can be seen, and simplicity then favors the shortest account that produces them. Explaining fine-tuning (the way the constants seem precisely set to permit life) quantitatively takes a candidate space, a measure and an observer weighting. Simplicity enters that comparison; it does not by itself prove that our constants are the shortest compatible values. The prediction is that the complete laws lie near the shortest description that can produce observers.

4.4 The hard problem of consciousness

David Chalmers's hard problem of consciousness (Chalmers, 1996), the question of why physical processes give rise to felt experience, assumes its own conclusion: it takes experience to be something other than structure, then finds that no structural explanation reaches it. Every candidate answer must specify determinate properties and relations, so a critic who excludes structure from constituting experience in advance has excluded every possible answer. Mind and body act on each other, so they share one structure, and any account of what an extra ingredient adds would itself be structure.

Sentience is felt transitions in modeling systems. Reflective consciousness is a transition in a system that models its own activity. The threshold and the mapping from organization to particular experiences remain research questions. A thing is its complete structure, so two systems cannot differ in experience without differing in structure.

4.5 Many-to-one as structural root

Many-to-one maps, maps that send different inputs to the same output, are the proposed common root of the Principle of Least Action (roughly, the rule that motion minimizes a quantity called action), coarse-grained entropy (a measure of how many detailed arrangements fit one coarse description) and emergence. Together they suggest one research program connecting microscopic structure to observable physics, and their mechanisms still have to be told apart. Stationary phase (an approximation where most paths cancel, leaving those near the classical path) can recover classical action equations, coarse graining groups microstates (exact detailed configurations of a system), and boundary conditions (constraints fixed at the start or edges of a system) and statistical assumptions can support an entropy gradient; none of these follows from many-to-one maps alone. Time's direction does not come from them either: exact evolution, the precise change of a state over time, is one-to-one, and Section 4.2 gives the reason time runs forward.

Points of zero length do not add up to a length, so I argue for a discrete physical account of extension, the way things take up space. Stated precisely, a sum of singleton lengths, the lengths of single points, is zero, and analysis, the mathematics of limits and continuity, does not identify that sum with interval length. A short discrete rule is worth searching for when it can recover the required geometry, dynamics and observations. Discrete and continuous models can agree exactly on specified observables (measurable quantities) while differing on others, so the physical task is to recover observed distances and dynamics from specified discrete relations.

5. Falsification

Matheism has two halves, and each could fail in its own way. The ontology (the account of what exists) is necessary, as 2 + 2 = 4 is, so no observation confirms it and one counterexample would refute it: anyone who describes a coherent feature that is neither ruled nor random, in physics or in experience, refutes the dichotomy and mathematical monism with it. Necessity leaves the claim its teeth. It rules out a bare substrate, a soul that moves a body by no law, a free choice that is neither caused nor random and a creator outside mathematics, each a position with defenders. The measure is the part that meets data: if the final theory of physics requires irreducibly high complexity, the Simplicity Dominance Principle is challenged, and the ontology stands either way. Both tests are stated in advance.

Stated precisely, falsifiability is a test for contingent empirical claims (ones that could have been otherwise), and a necessary claim is tested by counterexample instead. The partition is logic. That both branches are mathematical is mathematics' own result, since a rule is a function between states and randomness is algorithmic incompressibility or a probability law. That reality is identical to its complete structure has its own argument, from shared relations and an idle substrate (Section 1.4). The charge that the conclusion is empty fails on the list above: a claim that excludes those positions has content.

Conclusion

Laws, matter and minds are mathematical structure, because everything that exists is ruled or random, and both are mathematical. The central argument asks one question: once every property and relation of a thing is listed, what could a further substance add? The claim is necessary, as 2 + 2 = 4 is, and it has teeth: it rules out a bare substrate, a soul that moves a body by no law, a free choice that is neither caused nor random and a creator outside mathematics, and one

coherent feature that is neither ruled nor random would refute it. The same argument reaches mind. The hard problem of consciousness assumes its conclusion, and since a structure is one wherever it occurs, the bare subject of experience is one self in every mind, with distinct memories and lives left in place.

The physical work is to identify the structure of our universe, and the measure is the part that meets data. A shorter complete description receives greater initial weight, because each of its bits is an unruled 50/50 choice, and since whether to add another bit is a second 50/50, longer descriptions lose in total as well. The universe starts simple and grows the minds that observe it: no brains, no records and no instruments at the start. Growth runs in causal steps, because a cause produces its effect as the next state and a continuum has none, and time is those steps, running forward from the simple start toward minds that remember the past because they were built by it. To explain our particular laws, our quantum observations and the direction of time, one model has to deliver all three together.

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